TEK4710 Exercise 2

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1 Satellite imaging systems

The smartphone I use is the Samsung Galaxy A52 model.

1.1 a) Horizontal pixel number on a smartphone

After taking a picture, I use Google Photo to check the information. There it shows a combination of numbers

$$3468 \times 4624$$

which is the pixel resolution of the image. The pixel resolution is defined by $W \times H$, where W is the width and H the height. That means the number of horizontal pixels is 3468.

1.2 b) Field of view (degrees) on smartphone

From Exercise 1 we already specified the physical focal length and f-number (found in Google Photos app).

$$\text{physical focal length} = 5.23 \text{ mm}$$

The other thing we need is the sensor dimension (in this case, the sensor height would be nice). The sensor contains small plates with physical pixels that capture an image. To calculate the physical sensor height, we need the number of vertical pixel, which is just $H = 4624$. It's specified on the phone that each pixel size is around 0.8μm. This means that the sensor height h is

$$h = H \cdot 0.8\mu\text{m} = 3.70 \text{ mm}$$

Thus, the angular field of view on the phone can be approximated as illustrated below:
Insert illustration

The vertical field of view θ_H can be expressed as

$$\begin{aligned}\theta_H &= 2 \cdot \arctan\Big(\frac{H \cdot 0.8\mu\text{m}}{2 \cdot \text{focal length}}\Big) \\ &= 2 \cdot \theta \\ &= 38.96 \text{ degrees}\end{aligned}$$

We must also consider the horizontal field of view θ_W, expressed as

$$\begin{aligned}\theta_W &= 2 \cdot \arctan\Big(\frac{W \cdot 0.8\mu\text{m}}{2 \cdot \text{focal length}}\Big) \\ &= 29.67 \text{ degrees}\end{aligned}$$

Which mean that the vertical field of view is 38.96 degrees and the horizontal field of view is 29.67 degrees.

1.3 c) Smartphone pixel size from satellite orbit

The satellite orbits at around $L = 600$ km above ground, so we can assume the ground is flat when taking a picture with the PhoneSat setup.

The ground area captured by the satellite given its horizontal and vertical field of view. This is illustrated below.

The area on the ground A_G can be found by its width w_G and length h_G

$$A_G = 4L^2 \tan(\theta_W/2) \tan(\theta_H/2)$$

and the area A_G is covered by 3468×4624 pixels, which means that the area covered by each single pixel is

$$\frac{A_G}{3468 \times 4624} = 8413 \text{ m}^2/\text{pixel}$$

1.4 d) Size of detector pixel on image sensor

The size of the detector pixel_1 on the image sensor is 0.8μm.

1.5 e) Smartphone exposure time in satellite orbit

Exposure time (also called shutter speed) is the amount of time the camera is open and takes in light. The satellite ground speed is $v = 7$ km/s. The exposure time equal to the time the satellite moves one pixel is the time it takes for the satellite to travel one pixel length at ground level.

From **d)** we know that the area covered by a single pixel at ground level is 8413 m^2, and since the pixel length are the same on each side, we get the pixel length

$$\sqrt{8413 \text{ m}^2} = 91.7 \text{ m}$$

The exposure time t_e is

$$\begin{aligned} t_e &= \frac{\text{ground pixel size}}{v} \\ &= \frac{91.7 \text{ m}}{v} = 0.0131 \text{ s} \end{aligned}$$

Which is fairly reasonable, as the exposure time on my own phone is at $1/50$ s $= 0.02$ s.

1.6 f) Hubble pixel size from satellite orbit

Given that the Hubble's sensor pixel is 15 μm and the overall (effective) focal length is 57.6 m, we can use some geometry relations to see that the ground pixel size is

$$\begin{aligned} \frac{\text{ground pixel size}}{L} &= \frac{\text{sensor pixel size}}{\text{focal length}} \\ \text{ground pixel size} &= \frac{L \cdot \text{sensor pixel size}}{\text{focal length}} \\ \text{ground pixel size} &= 0.156 \text{ m} \end{aligned}$$

Since we only look at one ground pixel, it is not necessary to calculate the whole area coverage.

1.7 g) Hubble exposure time in satellite orbit

Assuming its similar conditions as in **e)**, then the exposure time for Hubble is

$$\begin{aligned} t_e &= \frac{\text{ground pixel size}}{v} \\ &= \frac{0.156 \text{ m}}{v} = 2.23 \cdot 10^{-5} \text{ s} \end{aligned}$$

Which mean that the shutter speed of Hubble when pointed nadir is $2.23 \cdot 10^{-5}$ s.

1.8 h) Frame rate: phone vs Hubble

Frame rate is the number of images recorded per second. We assume in both the phone and Hubble case that each recorded image have a 50% overlap. That means that after capturing an image it has moved 50 % of N_Y before capturing the next image. For our phone, $N_Y = 3468$ pixels. We know from **e)** and **g)** that the travel speed per ground pixel size is equal to the shutter speed, which means that the frame rate is

$$\begin{aligned} \text{fps} &= \frac{1}{\text{time between image and next image}} \\ &= \frac{1}{50\% \cdot N_Y \cdot \text{travel speed per ground pixel}} \end{aligned}$$

For the phone, the frame rate is

$$\text{fps} = \frac{1}{50\% \cdot 3468 \cdot 0.0131 \text{ s/pixel}} = 0.044 \text{ frames/s}$$

and for Hubble, the frame rate is

$$\text{fps} = \frac{1}{50\% \cdot 4096 \cdot 2.23 \cdot 10^{-5} \text{ s/pixel}} = 21.882 \text{ frames/s}$$

1.9 i) Data rate smartphone

The data rate is amount of data generated per second. This can be expressed as

$$\text{data rate} = (\text{data/pixel}) \cdot (\text{pixels/image}) \cdot (\text{frame rate})$$

The phone uses 3 bytes of data per pixel (RGB color). Thus, the data rate is

$$\begin{aligned}\text{data rate} &= (3 \text{ bytes/pixel}) \cdot (3468 \cdot 4624 \text{ pixels/image}) \cdot (0.044 \text{ images/s}) \\ &= 2116756 \text{ bytes/s} \\ &= 2.11 \text{ MB/s}\end{aligned}$$

1.10 j) Data rate Hubble

Hubble uses 14 bits of data per pixel (gray scale). Thus, the data rate is

$$\begin{aligned}\text{data rate} &= (14 \text{ bits/pixel}) \cdot (4096^2 \text{ pixels/image}) \cdot (21.882 \text{ images/s}) \\ &= 5139666567 \text{ bits/s} \\ &= 642458321 \text{ bytes/s} \\ &= 642 \text{ MB/s}\end{aligned}$$

since 1 byte = 8 bits.

1.11 k) Snow area and green band radiance received by satellite

We consider a white-snow covered surface with 100 % diffuse reflectance, meaning that the reflected light scattering is uniform. The solar spectral irradiance is 1.3 W/(m^2 nm). For simplicity, lets first just ignore the wavelength scale (will add it back later). Then the radiance becomes 1.3 W/m^2 since its constant on the wavelength interval. For simplicity, lets assumed 100 % reflectance of the solar irradiance. And atmospheric absorption is neglected. We are assuming the satellite is the PhoneSat, in which case (assuming the whole area covered by the phone camera is white snow) the area of the source is

$$\begin{aligned}A_{\text{source}} &= (\text{ground area per pixel}) \cdot (\text{number of pixels}) \\ &= 8413 \text{ m}^2/\text{pixel} \cdot 3468 \cdot 4264 \\ &= 1.35 \cdot 10^{11} \text{ m}^2\end{aligned}$$

and the radiance from the snow per area is $L_{\text{source}} = 1.3 \ W/\text{m}^2$. Then the intensity I is

$$\begin{aligned}I &= L_{\text{source}} \cdot A_{\text{source}} \\ &= 1.755 \cdot 10^{11} \ W\end{aligned}$$

And furthermore, we can assume the white-snow area being a large open field. The solid angle subtended by the sky is 2π sr. This means that the intensity I have a solid angle subtended by the sky which is 2π sr. Thus, the radiant intensity per steradians (sr) is

$$I_e = \frac{I}{2\pi \text{ sr}} = 2.79 \cdot 10^{10} \text{ W/sr}$$

The satellite (PhoneSat) has a $10 \times 10 \times 10$ cm^3, but lets assume the projected area is 10×10 cm. The satellite orbits at $d = 600$ km above ground. Then the solid angle subtended by the satellite (PhoneSat) can be approximated using

$$\begin{aligned}d\Omega &\approx \frac{\text{satellite projected area}}{d^2} \\ &= \frac{0.01 \text{ m}^2}{600000^2 \text{ m}^2} = 2.78 \cdot 10^{-14} \text{ sr}\end{aligned}$$

Then the radiance the satellite receives on it's projected area of 0.01 m^2 is

$$I_e \cdot d\Omega = 7.756 \cdot 10^{-4} \text{ W}$$

and since we want the radiance received by the satellite in W/m^2 and we have to consider the green light wavelength scale, we get

$$E_e = \frac{I_e \cdot d\Omega}{\text{satellite projected area}} \cdot \text{green light wavelength}$$

$$= 0.0775 \ \frac{W}{\text{m}^2 \text{ nm}} \cdot 525 \text{ nm}$$

$$= 40.7 \ W/\text{m}^2$$

Thus, the radiance received by the satellite on the green band is 40.7 W/m^2.

1.12 l) Snow area photoelectron count of phone detector pixel

We know the following for the PhoneSat:

Focal length: 5.23 mm, focal number: 1.8 $\longrightarrow$ Aperture diameter $d = 2.91$ mm $\longrightarrow$ Lens area $A_d = 6.65 \cdot 10^{-6}$ m^2

Integration time (exposure time): $t_{\text{int}} = 0.0131$ s

Pixel size: 0.8 μm $\longrightarrow$ $6.4 \cdot 10^{-13}$ m^2/detector pixel $\longrightarrow$ Detector area $A_D = 1.026 \cdot 10^{-5}$ m^2

Detector resolution: 3468×4624

Radiance received by the satellite on the snow area on the green band ($\lambda = 525$ nm): $E_e = 40.7$ $\mathrm{W/m}^2$

First we calculate the photon flux received by the lens. The lens distributes the same photon flux to the detector pixels. We can then measure the photon flux a single detector pixel receives. Then we convert the photon flux into photoelectron count for the given wavelength and quantum efficiency and integration time.

We can measure the photon flux ϕ_{q} using the received radiance E_e and energy of a photon of a certain wavelength E

$$\begin{aligned}\phi_{\text{q}} &= \frac{E_e}{E} \\ &= \frac{E_e \cdot \lambda}{hc} \\ &= 1.056 \cdot 10^{20} \text{ photons/m}^2\text{s}\end{aligned}$$

where $h = 6.626 \cdot 10^{-34}$ Js is the Planck's constant, while c is the speed of light.

Now, the photon flux received by the lens is

$$\begin{aligned}\phi_d &= A_d \cdot \phi_{\text{q}} \\ &= 7.0224 \cdot 10^{14} \text{ photons/s}\end{aligned}$$

and that is the same photon flux the detector pixels receive. For a single detector pixel, the photon flux it receives is

$$\begin{aligned}\phi_{\text{pix}} &= \frac{\phi_d}{\text{pixel resolution}} \\ &= 4.38 \cdot 10^7 \text{ photons/s}\end{aligned}$$

thus, the photoelectron count n_e is

$$\begin{aligned}n_e &= \int_{t_{\text{int}}} \eta(\lambda) \cdot \phi_{\text{pix}} \, dt \\ &= \frac{1}{2} \cdot \phi_{\text{pix}} \int_{t_{\text{int}}} dt \\ &= \frac{1}{2} \cdot \phi_{\text{pix}} \cdot t_{\text{int}} \\ &= 286890\end{aligned}$$

1.13 n) Snow area photoelectron count of Hubble

We know the following for Hubble:

Focal length (effective): 57.6 m, Aperture diameter $d = 2.4$ m $\longrightarrow$ Lens area $A_d = 4.524$ m^2

Integration time (exposure time): $t_{\text{int}} = 2.23 \cdot 10^{-5}$ s

Pixel size: 15 μ $\longrightarrow$ Detector area $A_D = 3.775 \cdot 10^{-3}$ m^2

Detector resolution: 4096×4096

The radiance received by Hubble is the same as the one received by the satellite on the green band: E_e

Using the same formulation as for the PhoneSat, we can simplify the formula for the photoelectron count to be

$$\begin{aligned} n_e &= \frac{t_{\text{int}} \cdot A_d \cdot E_e \cdot \lambda}{2 \cdot h \cdot c \cdot (\text{pixel resolution})} \\ &= 3.23 \cdot 10^8 \end{aligned}$$

2 Investigate limitations of a camera

From my phone, I use the Siemens star pattern.

2.1 a) Resolution

Using my phone which has an original resolution of 3468×4624, I was able to maximally zoom in on the center of the Siemens star image down to a resolution of 224×224. Then, using Python (and importing packages like PIL), I was able to plot the image where the smallest detail is the black and white edge squares (as shown in **Figure 1**). Zooming in on this detail shows that the black square covers around 6×6 pixels.

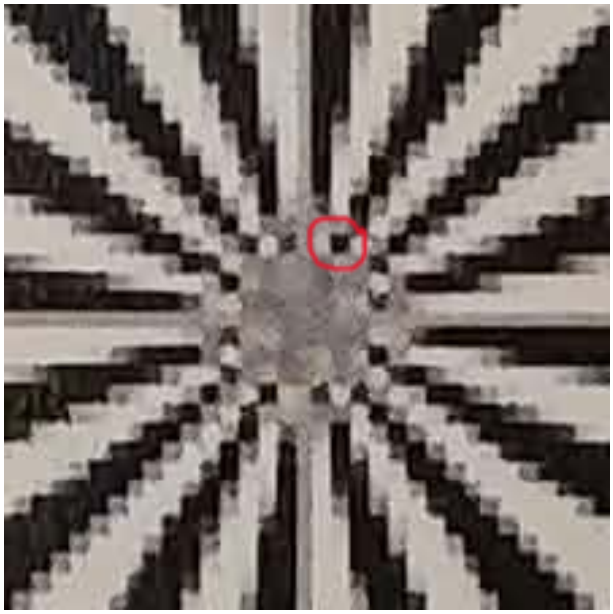


Figure 1: Siemens star pattern photo taken from my phone and cropped with maximum zoom (resolution of 224×224). The red mark indicates the smallest clear detail. Any details further towards the center are blurred

The angle seen from the camera onto that smallest detail can be approximated using the phones physical focal length $F = 5.23$ mm and the amount of detector pixels that small detail takes. Using the fact that each pixel has a size of 0.8 μm, we can express the angle as

$$\begin{aligned} \theta &= 2 \cdot \arctan\Big(\frac{6 \text{ pixel size}}{2F}\Big) \\ &= 2 \cdot \arctan\Big(\frac{0.0048 \text{ mm}}{2 \cdot 5.23 \text{ mm}}\Big) \\ &\approx 0.0526^\circ \end{aligned}$$

2.2 c) Saturation

We have 3 photos of the lamp to measure the saturation under different exposure times (**Figure 2**). Using the phone camera's built-in Pro-mode we can control the exposure time. The pictures are in RGB format before processing, and after processing it becomes binary format. We have two versions; saturated and white.

Saturated are all the pixels that have at least one of the color channels at maximum value (255).

White are all the pixels that have maximum value on all the color channels (as white light contains all the possible visible wavelengths).

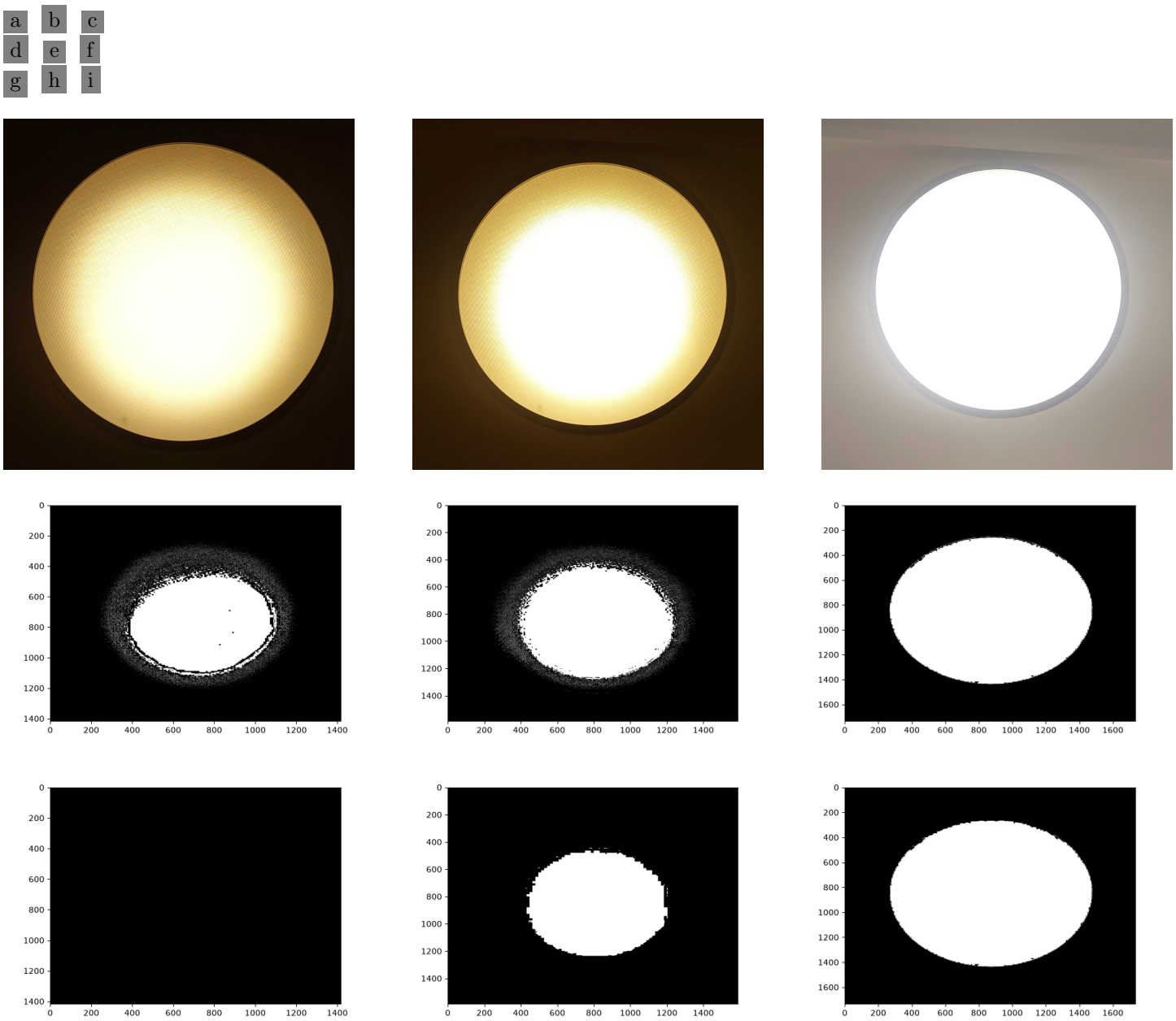


Figure 2: Results of saturation based on original photo, using the same ISO50. **a-c**: The original photos taken with respective exposure times of 1/1000s, 1/500s and 1/50s. **d-f**: The saturated parts with respective exposure times of 1/1000s, 1/500s and 1/50s. **g-i**: The fully saturated parts (white) with respective exposure times of 1/1000s, 1/500s and 1/50s.

We can observe as the exposure times increases, the amount of saturated/white pixel areas appears more. Due to the fact that the detector pixels receives more photoelectrons count when exposure time is longer, the resulting digital image can intentionally become more saturated. An example of this is taking pictures of Auroras with prolonged exposure time (using Pro-mode on Android phones). The picture will appear more "vibrant and clear", with the features of the Aurora and the night sky becoming "brighter" than when seen with the eye (examples shown below).

2.3 d) Integration time

We record a picture of a moving object while the background is in focus. Below (**Figure 3**), its illustrated with a background that have a drawn square to show clear background focus, while a marker is used as the object moving in that frame.

In these pictures there are two key camera features to consider: exposure time and ISO.

Exposure time is what controls how much light is let in and processed by the detector sensors. When the object in frame moves during the exposure time it causes the detector pixels covered by the moving object to be exposed to a sudden change in the wavelength they receive. This is what causes the big blur in the moving object. As observed with further experiments, when the exposure time is reduced, the area of pixels that get exposed by the moving object also gets smaller, which leads to a recorded image with less blur. But a smaller exposure time also means that the detector receives less light, i.e. less saturated. This is where the ISO-number comes in. The ISO-number essentially acts as a digital pixel amplifier that takes

low saturation and turns it into a higher saturated image. A higher ISO-number means the image gets a higher saturation.

a b c

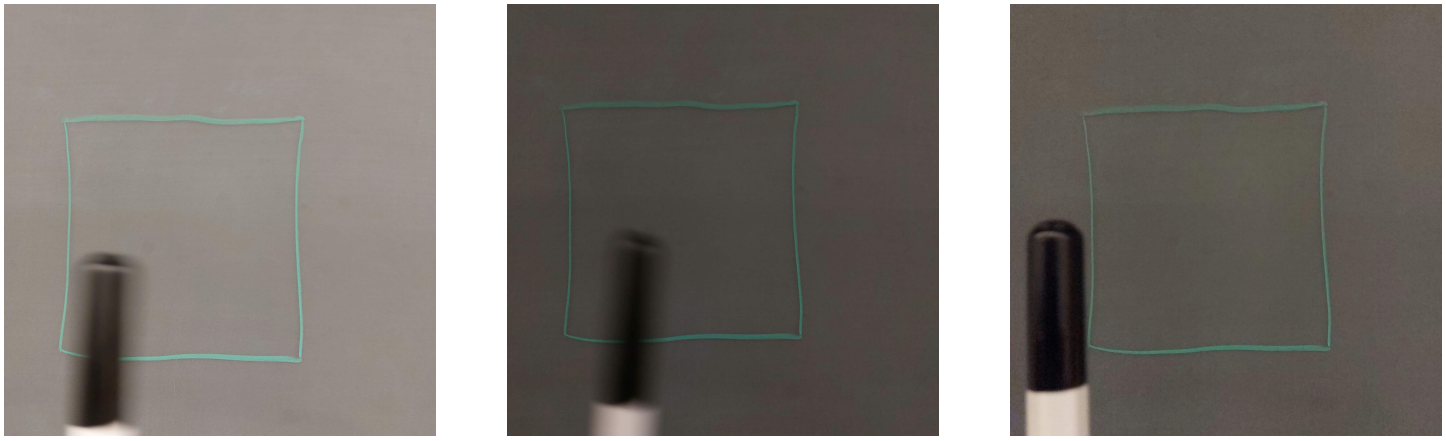


Figure 3: Photos of object moving (black marker) blur based on different exposure time. **a**: ISO320 and 1/50s. **b**: ISO200 and 1/90s. **c**: ISO1600 and 1/500s.

2.4 e) Focus

The Bokeh effect refers to the quality of the blur in the parts of the image that are out of focus[2]. We have an object in focus (the Portal Space cup) at different distances of roughly 0.2m, 0.4m and 1m. As observed, the background is very blurred for the close focus (0.2m), but gradually the background gets clearer as the object in focus gets further away (the paintings on the left side of the focus object).

a b c



Figure 4: Photos of object in focus (Portal Space cup) recorded at different distances to observe the Bokeh effect. **a**: 0.2m. **b**: 0.4m. **c**: 1m.

Since the object is further away, there is less focus on the object, which makes the background resolution better and thus less blurry. For reference, we can use Figure 68 from the lecture notes.

2.5 b) Low-light limit

Low-light limit refers to the case where the picture starts blurring due to the low amount of light received by the sensors. To illustrate this, the pictures were uploaded and plotted (in greyscale) using Python in order to enhance the visuals and also highlight the noise that occurs during low-light conditions (as observed in **Figure 5**) since ordinary phone camera pictures under very low light levels were hard to observe.

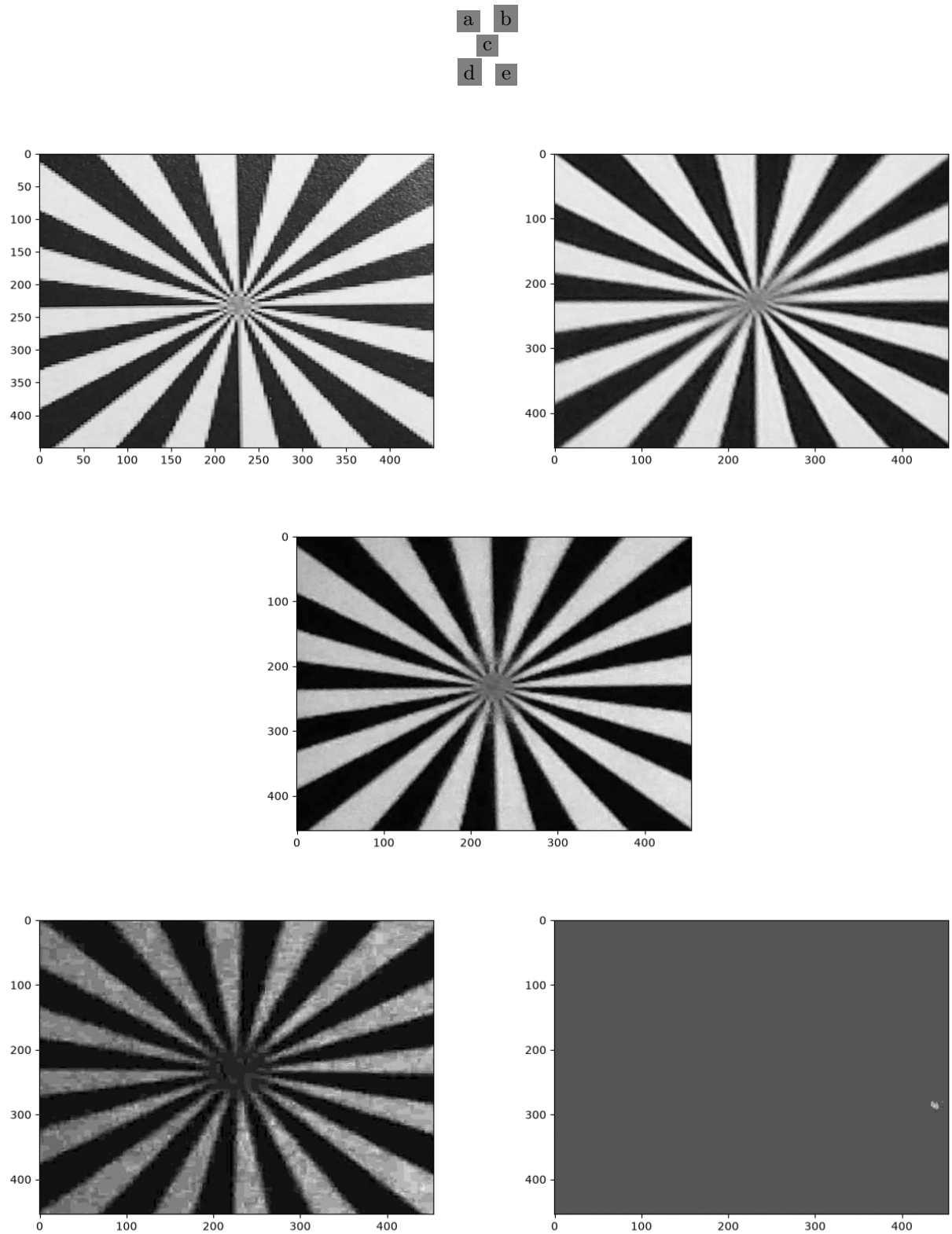


Figure 5: Low-light images of the same pattern to show increasing blur and noise when the light levels are decreasing. The images are taken from the phone using ordinary camera mode. **a**: Full light. **b**: Less light. **c**: Even less light. **d**: Almost no light. **e**: Complete darkness

We can observe from **Figure 5** that as the light levels decreases, the pattern details becomes more blurry and noise-apparent, except for the complete darkness. This blurring can be observed between **Figure 5: a-c**. In Figure 5**a** we can see the "staircases" on the edges of the black lines. For Figure 5**b** we are still able to observe some of the "staircases", although they are roughed out now. Moving on to Figure 5**c** the "staircases" are now blurred out, and in addition, there are some signs of noise appearing, especially when we look at the white parts. In Figure 5**d** we have very low light (the original picture on the phone appeared way darker than the plot), which makes the pattern a bit more blurred, but now the photon noise is also very apparent. Lastly, in Figure 5**e** the noise seemingly have disappeared with the exception of a very tiny region on the middle-right, which can be speculated to be phone light affecting the surroundings or other effects.

The photons that hit the detector sensors are somewhat controlled by statistical distribution patterns (like normal distribution, uniform, etc). These distribution patterns become clearer when more photons are involved. As the light levels decrease, the statistical distribution becomes more rough, which leads to these noises in the images.

2.6 f) Uniformity

We take a photo of a blank piece of paper under normal brightness, as the paper is a considerable smooth, flat and uniform surface (all pixels on the surface appears the same). To adjust and "maximize" the contrast, some Python tools were utilized from the course **IN3370**. In short, we can maximize and adjust the contrast of the image by calculating the total average μ of the pixel values. The results can be seen in **Figure 6**.

a b c
d e f

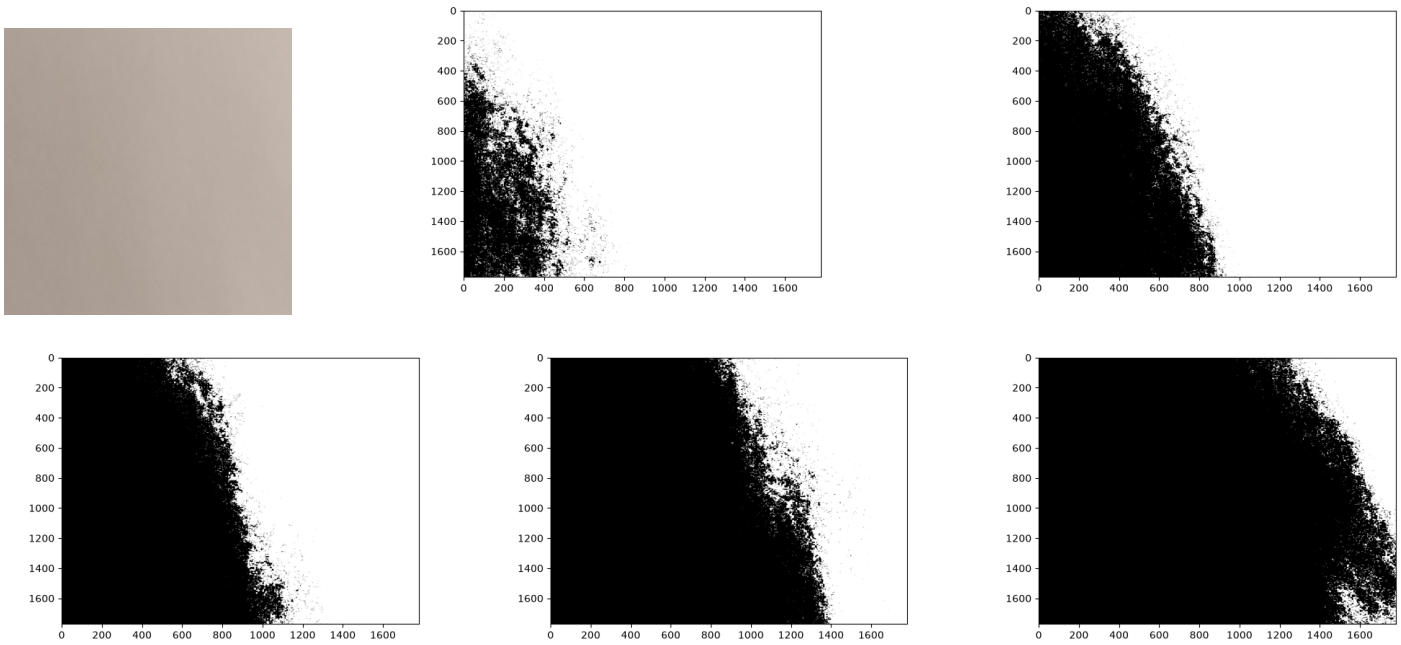


Figure 6: Original uniform picture and adjusted contrast values to highlight a fixed noise pattern using $\mu = 169$ as a threshold benchmark. **a**: Original picture. **b**: Contrast with $\mu - 10$. **c**: Contrast with $\mu - 5$. **d**: Contrast with μ. **e**: Contrast with $\mu + 5$. **f**: Contrast with $\mu + 10$.

As we can observe, although the picture appears as uniform before processing, after processing it with different values of threshold μ, we can see that the noise pattern edge seems to move in a diagonal shearing way.

The threshold value here is calculated to be $\mu = 169$, which is the total average of all the pixels in the original image. Then we iterate through each pixel in the original image and set it to black (0) if it is below the threshold μ, and to white (1 or 255) otherwise. To show the fixed noise pattern, we adjust the value of μ with some small perturbations, as indicated in **Figure 6**. The five different values of the threshold was [159, 164, 169, 174, 179].

3 References and tools

All the plots for **Figure 2, Figure 5** and **Figure 6** can be found under the codefile **tek_exercise_2.py**.

3.1 Sources

[1] Phone specifications: https://www.gsmarena.com/samsung_galaxy_a52_5g-10631.php

[2] Bokah effect: https://www.youtube.com/shorts/ReThY2a8BCk

3.2 Code availability

Code for this exercise is available at:

https://github.com/zwl-2012/TEK4710/tree/main/Assigments/Exercise%202